

# Serverless Student Performance Analytics System

## Solution Design Document (SDD)

**Project Name:** Serverless Student Performance Analytics System

**Prepared By:** Ayush Agrawal

### Technology Stack:

- AWS S3
- AWS Lambda (Python)
- Amazon DynamoDB
- AWS IAM
- CloudWatch Logs
- Boto3 SDK

## 1. Executive Summary

The Serverless Student Performance Analytics System is designed to automate the ingestion, processing, storage, and analysis of student academic performance data.

The solution leverages AWS serverless services to eliminate infrastructure management while providing scalability, reliability, and cost efficiency.

Faculty members upload student performance datasets into Amazon S3. Upon upload, an AWS Lambda function is automatically triggered to process the data, calculate performance metrics, and store records in DynamoDB.

The system supports:

- Automated data ingestion
- CRUD operations
- Performance analysis
- Student risk detection

- Leaderboard generation
- Data export functionality
- Advanced querying using DynamoDB Global Secondary Indexes (GSI)

## **2. Business Problem**

Universities collect student performance data every week.

Challenges:

- Manual data entry
- Lack of centralized storage
- Delayed performance analysis
- Difficulty identifying at-risk students
- No automated reporting

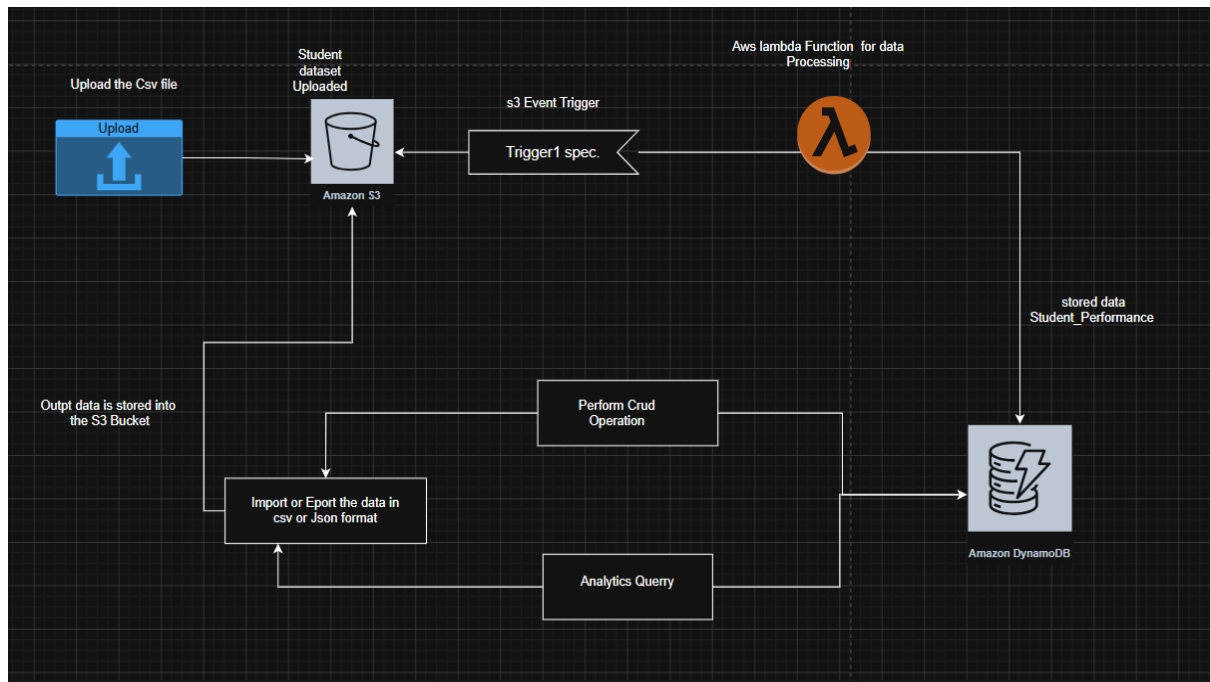
The proposed solution automates the complete data pipeline.

## **3. Objectives**

The system aims to:

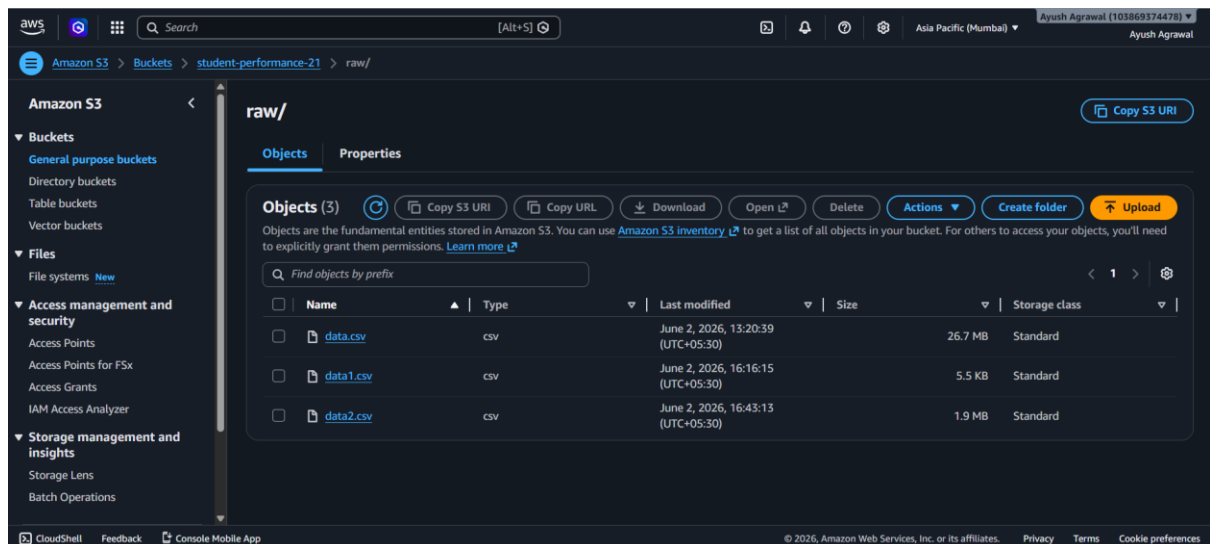
- ✓ Automate student data ingestion
- ✓ Store student records efficiently
- ✓ Support real-time analytics
- ✓ Identify high-performing students
- ✓ Detect students at academic risk
- ✓ Generate performance reports
- ✓ Reduce operational overhead through serverless architecture

## 4. High-Level Architecture

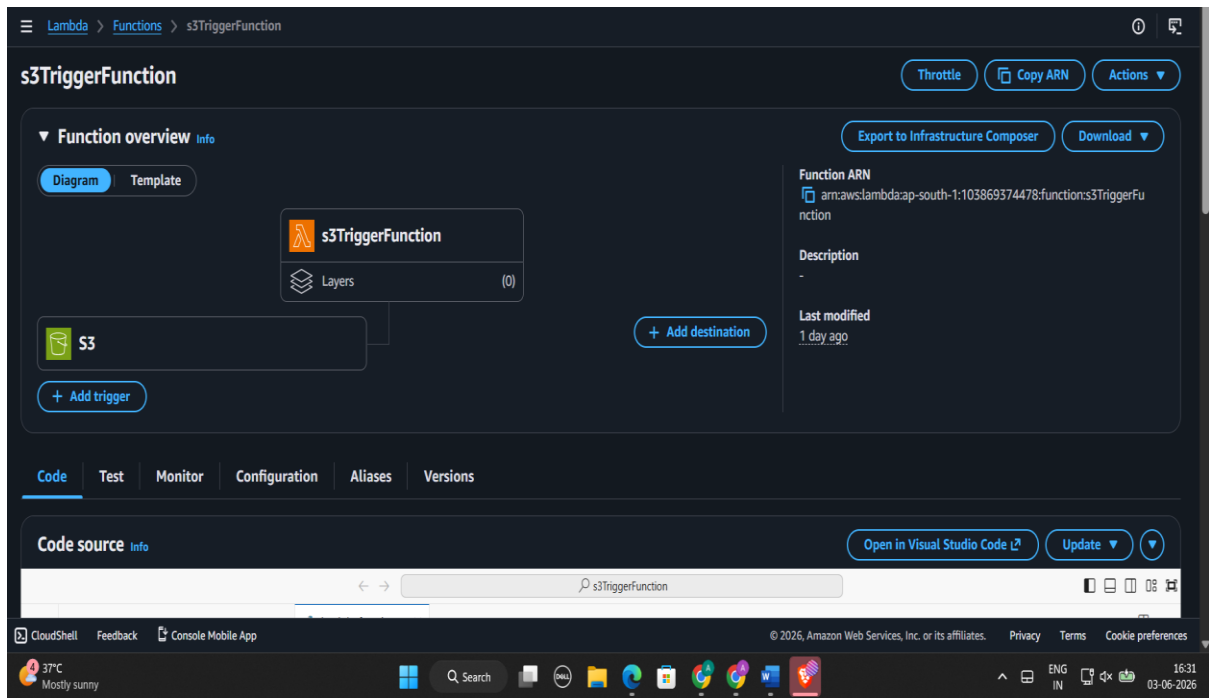


## 5. AWS Services Used

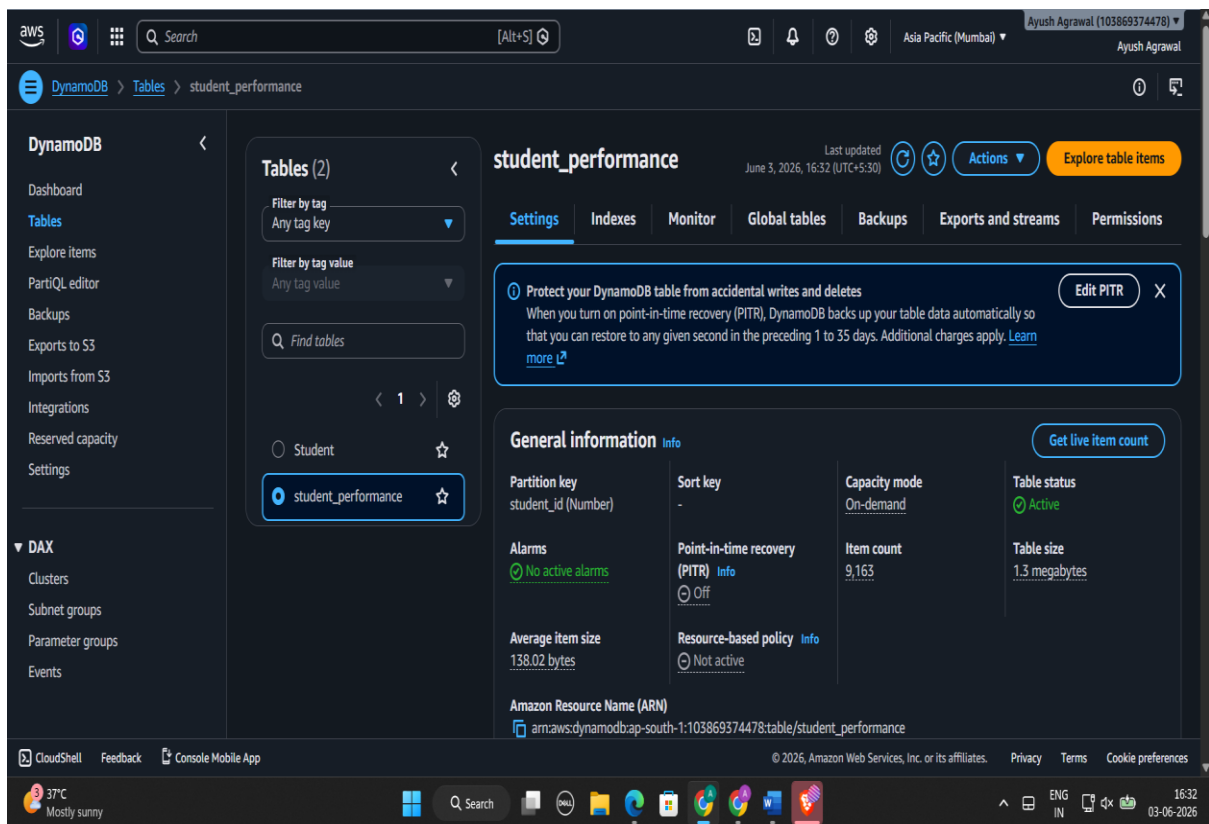
### Aws s3 storage



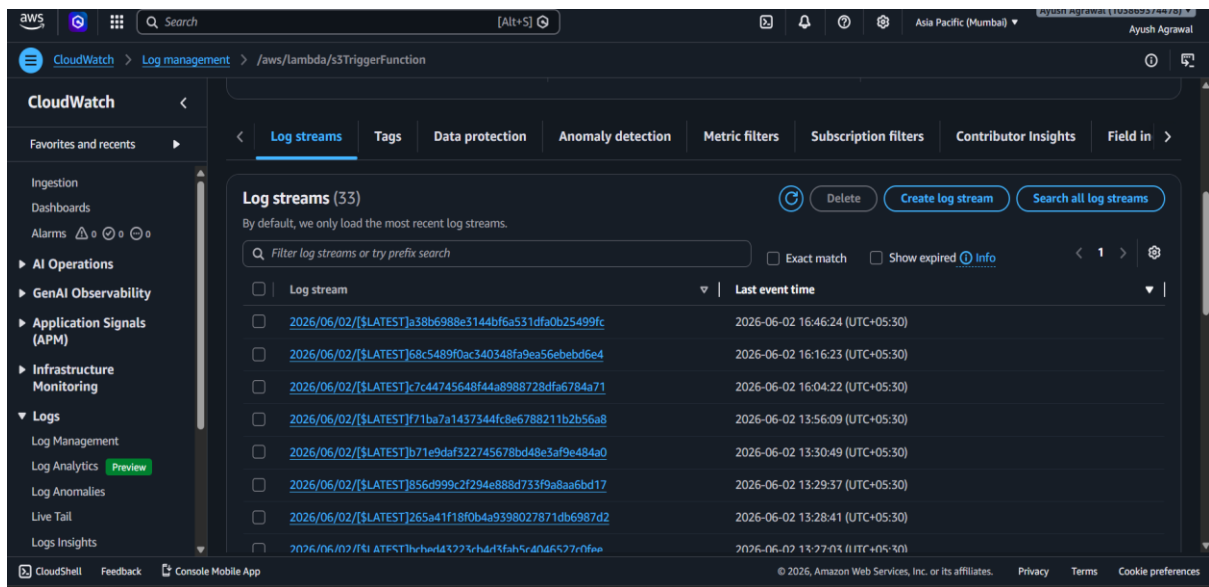
### AWS Lambda



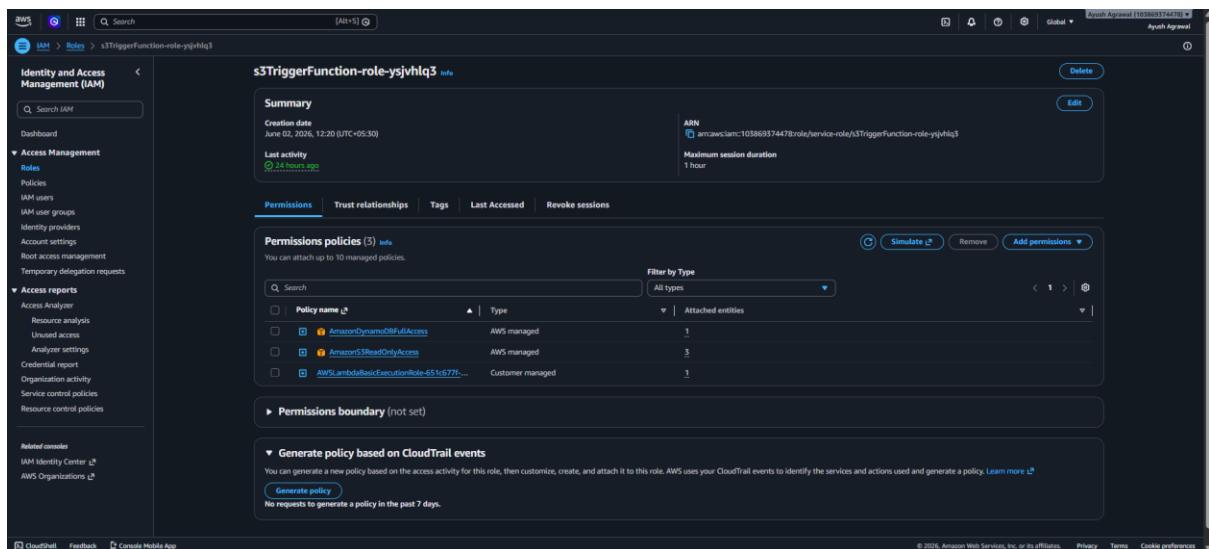
## DynamoDb



# CloudWatch



# IAM role



All the policies is define to the lambda function means I give the access to the lambda function of s3ReadOnly, DynamoDb and AWSLambdaExecutionRole

## DynamoDb Attribute

## Primary Key

- Partition-Key: Student\_ ID

## Secondary-Key(GSI)

- Partition-key: Student\_Id
- Sort-key: Total\_score

## Attributes

- Student\_id
- weekly\_self\_study\_hours
- Attendance\_Percentage
- Total\_Score
- Class\_Participation
- grade
- Performance\_Category

## Lambda Processing Workflow

Start

|

Read S3 File

|

Validate JSON

|

Loop Through Records

|

Calculate Category

|

Calculate Risk

|

Insert Into DynamoDB

|

Log Success(End)

